Performance Estimation Problems (PEPs)

Systematic, principled, and computer-aided approaches to the analysis and design of first-order optimization algorithms

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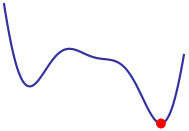
SMAI-MODE tutorial – March 2026

Context: numerical (continuous) optimization

Minimize $f : \mathbb{R}^d \to \mathbb{R}$ (e.g., with f continuous)

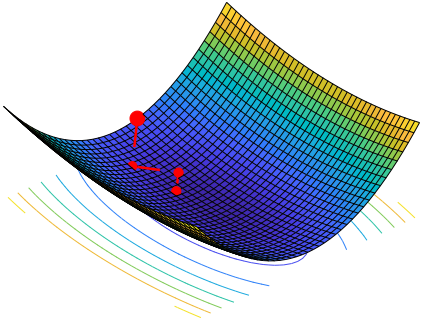
$$f(x_\star) \triangleq \min_{x \in \mathbb{R}^d} f(x).$$



Ubiquitous in applied mathematics and computer science.

Numerous applications for modeling (physics, economics), estimation (statistics, machine learning), decisions (control, operations research).

Usually solved via **iterative algorithm** generating sequence $x_0, x_1, \ldots, x_N$.



Gradient descent (stepsize α)

for $k = 0, 1, \ldots$ **do**

$$x_{k+1} = x_k - \alpha \nabla f(x_k)$$

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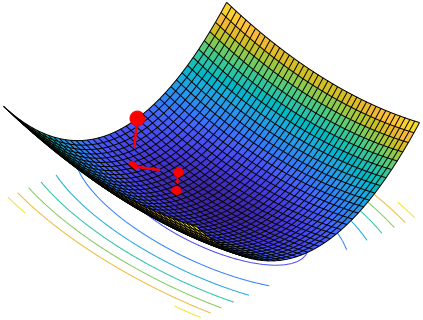

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For instance: **bounds** on certain notions of "error": $f(x_k) - f(x_\star)$, $\|x_k - x_\star\|$, $\|\nabla f(x_k)\|$, etc.

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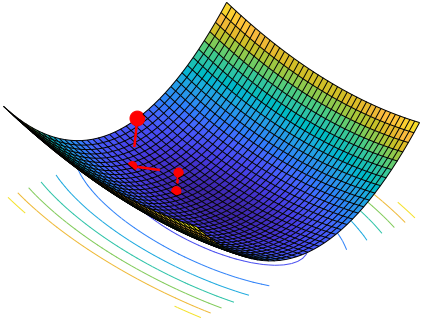

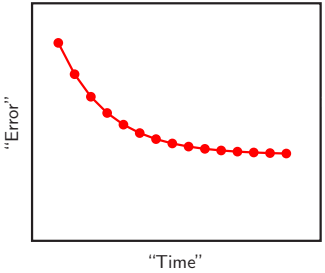

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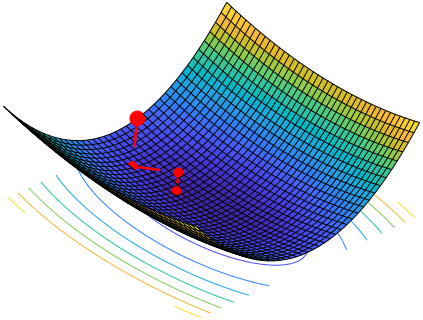

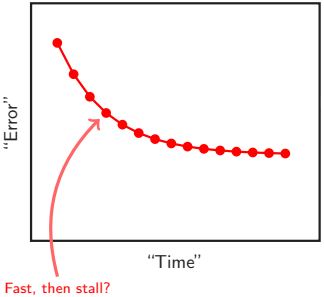

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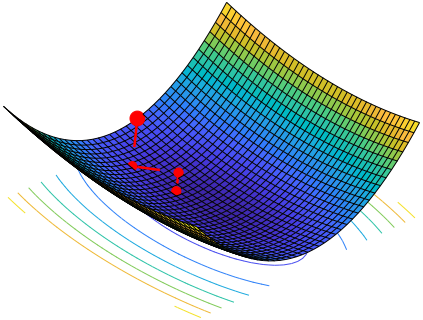

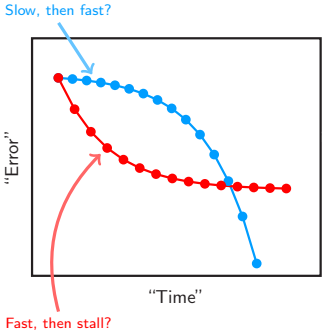

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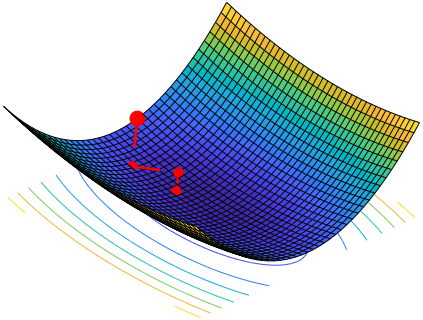

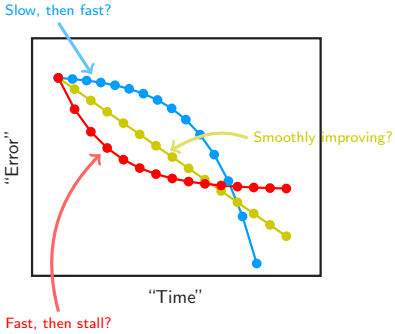

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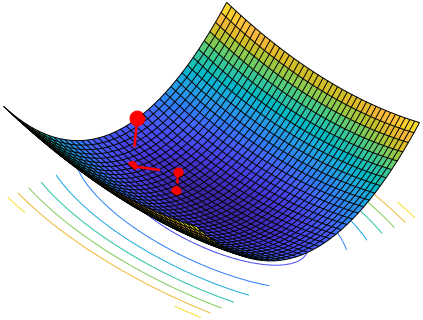

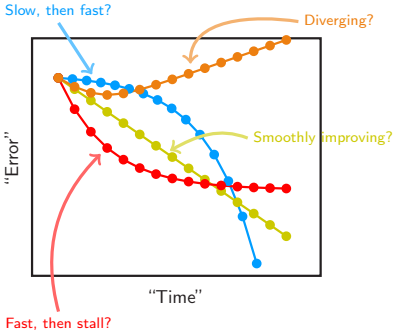

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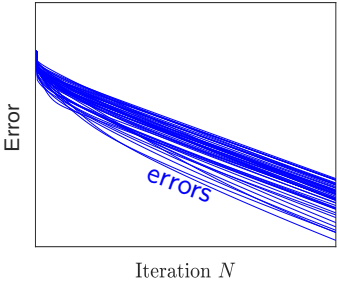

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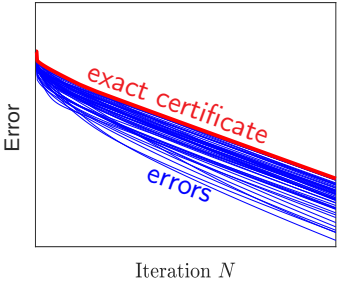

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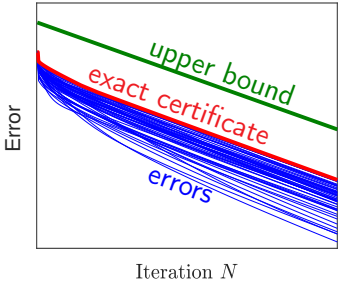

Constructive approach to performance analysis

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Example: analysis of a gradient method

Let $f : \mathbb{R}^d \to \mathbb{R}$ (continuously differentiable). Find $x_\star \in \mathbb{R}^d$ such that

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Question: what *a priori* guarantees after N iterations?

Examples: what about $f(x_N) - f(x_\star)$, $\|\nabla f(x_N)\|$, $\|x_N - x_\star\|$?

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A differentiable function $f : \mathbb{R}^d \to \mathbb{R}$ is μ-strongly convex and L-smooth iff $\forall x, y \in \mathbb{R}^d$:

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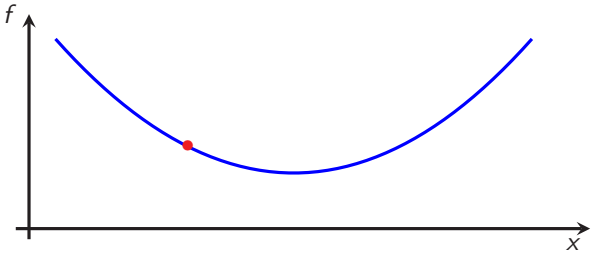

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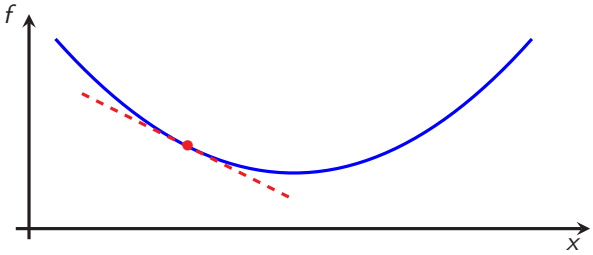

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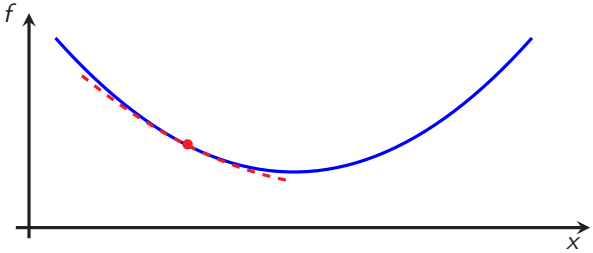

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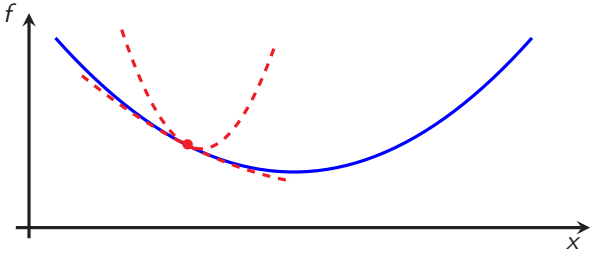

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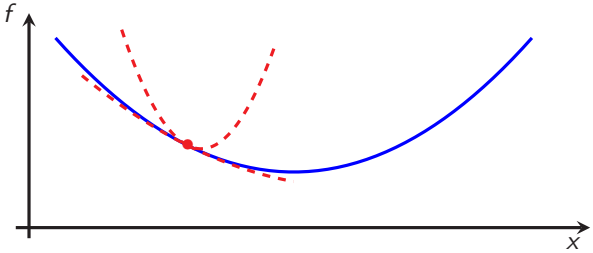

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(1&2) $\langle \nabla f(x) - \nabla f(y); x - y \rangle \geqslant \frac{1}{L+\mu}\|\nabla f(x) - \nabla f(y)\|^2 + \frac{\mu L}{L+\mu}\|x - y\|^2$.

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- Unique way to arrive to the desired result?
- How likely are we to find such proofs in more complicated cases?

Legitimate questions about performance analyses?

Lemma 3. *Assume that the function is L-smooth and μ strongly-convex and satisfies the strong-growth condition in Equation 10. Then, using the updates in Equation 3-5 and setting the parameters according to Equations 17-19, if $\eta \leq \frac{1}{\mu L}$, then the following relation holds:*

$$\mathbb{E}_{k+1}\gamma_k^2 \left[\mathbb{E}\|\nabla f(w_{k+1}) - f^*\right] \leq \frac{\alpha_k^2}{\rho\eta}\left(f(x_0) - f^*\right) + \frac{\eta_k^2}{2\rho\eta}\|x_0 - w^*\|^2 + \frac{\sigma^2\eta}{\rho}\sum_{t=0}^{k}\left[\gamma_t^2\eta_{t+1}^2\right]$$

Proof.

Let $r_{k+1} = \|w_{k+1} - w^*\|$, then using equation 5

$$r_{k+1}^2 = \|\beta_k v_k + (1-\beta_k)\zeta_k - w^* - \gamma_k\eta\nabla f(\zeta_k, z_k)\|^2$$
$$r_{k+1}^2 = \|\beta_k v_k + (1-\beta_k)\zeta_k - w^*\|^2 + \gamma_k^2\eta^2\|\nabla f(\zeta_k, z_k)\|^2 + 2\gamma_k\eta\langle w^* - \beta_k v_k - (1-\beta_k)\zeta_k, \nabla f(\zeta_k, z_k)\rangle$$

Taking expectation wrt to z_k,

$$\begin{aligned}
\mathbb{E}[r_{k+1}^2] &= \mathbb{E}\|\beta_k v_k + (1-\beta_k)\zeta_k - w^*\|^2 + \gamma_k^2\eta^2\mathbb{E}\|\nabla f(\zeta_k, z_k)\|^2 + 2\gamma_k\eta\mathbb{E}[\langle w^* - \beta_k v_k - (1-\beta_k)\zeta_k, \nabla f(\zeta_k, z_k)\rangle] \\
&\leq \|\beta_k v_k + (1-\beta_k)\zeta_k - w^*\|^2 + \gamma_k^2\eta^2\rho\|\nabla f(\zeta_k)\|^2 + 2\gamma_k\eta\langle w^* - \beta_k v_k - (1-\beta_k)\zeta_k, \nabla f(\zeta_k)\rangle + \gamma_k^2\eta^2\sigma^2 \\
&= \|\beta_k(v_k - w^*) + (1-\beta_k)(\zeta_k - w^*)\|^2 + \gamma_k^2\eta^2\rho\|\nabla f(\zeta_k)\|^2 + 2\gamma_k\eta\langle w^* - \beta_k v_k - (1-\beta_k)\zeta_k, \nabla f(\zeta_k)\rangle + \gamma_k^2\eta^2\sigma^2 \\
&\leq \beta_k\|v_k - w^*\|^2 + (1-\beta_k)\|\zeta_k - w^*\|^2 + \gamma_k^2\eta^2\rho\|\nabla f(\zeta_k)\|^2 + 2\gamma_k\eta\langle w^* - \beta_k v_k - (1-\beta_k)\zeta_k, \nabla f(\zeta_k)\rangle + \gamma_k^2\eta^2\sigma^2 \\
&\qquad \text{(By convexity of } \|\cdot\|^2\text{)} \\
&= \beta_k r_k^2 + (1-\beta_k)\|\zeta_k - w^*\|^2 + \gamma_k^2\eta^2\rho\|\nabla f(\zeta_k)\|^2 + 2\gamma_k\eta\langle w^* - \beta_k v_k - (1-\beta_k)\zeta_k, \nabla f(\zeta_k)\rangle + \gamma_k^2\eta^2\sigma^2 \\
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&= \beta_k r_k^2 + (1-\beta_k)\|\zeta_k - w^*\|^2 + \gamma_k^2\eta^2\rho\|\nabla f(\zeta_k)\|^2 + 2\gamma_k\eta\left\langle\frac{\beta_k(1-\alpha_k)}{\alpha_k}(w_k - \zeta_k) + w^* - \zeta_k, \nabla f(\zeta_k)\right\rangle + \gamma_k^2\eta^2\sigma^2 \\
&\qquad \text{(From equation 4)} \\
&= \beta_k r_k^2 + (1-\beta_k)\|\zeta_k - w^*\|^2 + \gamma_k^2\eta^2\rho\|\nabla f(\zeta_k)\|^2 + 2\gamma_k\eta\left[\frac{\beta_k(1-\alpha_k)}{\alpha_k}\langle\nabla f(\zeta_k), (w_k - \zeta_k)\rangle + \langle\nabla f(\zeta_k), w^* - \zeta_k\rangle\right] + \gamma_k^2\eta^2\sigma^2 \\
&\leq \beta_k r_k^2 + (1-\beta_k)\|\zeta_k - w^*\|^2 + \gamma_k^2\eta^2\rho\|\nabla f(\zeta_k)\|^2 + 2\gamma_k\eta\left[\frac{\beta_k(1-\alpha_k)}{\alpha_k}(f(w_k) - f(\zeta_k)) + \langle\nabla f(\zeta_k), w^* - \zeta_k\rangle\right] + \gamma_k^2\eta^2\sigma^2 \\
&\qquad \text{(By convexity)}
\end{aligned}$$

By strong convexity,

$$\begin{aligned}
\mathbb{E}[r_{k+1}^2] &\leq \beta_k r_k^2 + (1-\beta_k)\|\zeta_k - w^*\|^2 + \gamma_k^2\eta^2\rho\|\nabla f(\zeta_k)\|^2 \\
&+ 2\gamma_k\eta\left[\frac{\beta_k(1-\alpha_k)}{\alpha_k}(f(w_k) - f(\zeta_k)) + f^* - f(\zeta_k) - \frac{\mu}{2}\|\zeta_k - w^*\|^2\right] + \gamma_k^2\eta^2\sigma^2
\end{aligned} \tag{16}$$

By Lipschitz continuity of the gradient,

$$\begin{aligned}
f(w_{k+1}) - f(\zeta_k) &\leq \langle\nabla f(\zeta_k), w_{k+1} - \zeta_k\rangle + \frac{L}{2}\|w_{k+1} - \zeta_k\|^2 \\
&\leq -\eta\langle\nabla f(\zeta_k), \nabla f(\zeta_k, z_k)\rangle + \frac{L\eta^2}{2}\|\nabla f(\zeta_k, z_k)\|^2
\end{aligned}$$

Taking expectation wrt z_k and using equations 9, 10

$$\begin{aligned}
\mathbb{E}[f(w_{k+1}) - f(\zeta_k)] &\leq -\eta\|\nabla f(\zeta_k)\|^2 + \frac{L\rho\eta^2}{2}\|\nabla f(\zeta_k)\|^2 + \frac{L\eta^2\sigma^2}{2} \\
\mathbb{E}[f(w_{k+1}) - f(\zeta_k)] &\leq \left[-\eta + \frac{L\rho\eta^2}{2}\right]\|\nabla f(\zeta_k)\|^2 + \frac{L\eta^2\sigma^2}{2}
\end{aligned}$$

If $\eta \leq \frac{1}{\rho L}$,

$$\begin{aligned}
\mathbb{E}[f(w_{k+1}) - f(\zeta_k)] &\leq \left(\frac{-\eta}{2}\right)\|\nabla f(\zeta_k)\|^2 + \frac{L\eta^2\sigma^2}{2} \\
\implies \|\nabla f(\zeta_k)\|^2 &\leq \left(\frac{2}{\eta}\right)\mathbb{E}[f(\zeta_k) - f(w_{k+1})] + L\eta\sigma^2
\end{aligned} \tag{17}$$

From equations 16 and 17

$$\begin{aligned}
\mathbb{E}[r_{k+1}^2] &\leq \beta_k r_k^2 + (1-\beta_k)\|\zeta_k - w^*\|^2 + 2\gamma_k^2\eta\rho\mathbb{E}[f(\zeta_k) - f(w_{k+1})] \\
&+ 2\gamma_k\eta\left[\frac{\beta_k(1-\alpha_k)}{\alpha_k}(f(w_k) - f(\zeta_k)) + f^* - f(\zeta_k) - \frac{\mu}{2}\|\zeta_k - w^*\|^2\right] + \gamma_k^2\eta^2\sigma^2 + L\gamma_k^2\eta^2\rho\sigma^2 \\
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&+ 2\gamma_k\eta\left[\frac{\beta_k(1-\alpha_k)}{\alpha_k}(f(w_k) - f(\zeta_k)) + f^* - f(\zeta_k) - \frac{\mu}{2}\|\zeta_k - w^*\|^2\right] + 2\gamma_k^2\eta^2\sigma^2 \qquad \text{(Since } \eta \leq \tfrac{1}{\rho L}\text{)} \\
&= \beta_k r_k^2 + \|\zeta_k - w^*\|^2[(1-\beta_k) - \gamma_k\mu\eta] + f(\zeta_k)\left[2\gamma_k^2\eta\rho - 2\gamma_k\eta\cdot\frac{\beta_k(1-\alpha_k)}{\alpha_k} - 2\gamma_k\eta\right] \\
&- 2\gamma_k^2\eta\rho\mathbb{E}f(w_{k+1}) + 2\gamma_k\eta f^* + \left[2\gamma_k\eta\cdot\frac{\beta_k(1-\alpha_k)}{\alpha_k}\right]f(w_k) + 2\gamma_k^2\eta^2\sigma^2
\end{aligned}$$

Example - do not read!

Example - do not read!

Legitimate questions about performance analyses?

Lemma 3. *Assume that the function is L-smooth and μ strongly-convex and satisfies the strong-growth condition in Equation 10. Then, using the updates in Equation 3-5 and setting the parameters according to Equations 17-18, if $\eta \leq \frac{1}{\mu L}$, then the following relation holds:*

$$b_{k+1}^2 \eta \left[\mathbb{E}[f(w_{k+1}) - f^*]\right] \leq \frac{\alpha_k^2}{\rho\eta}\left(f(\alpha_k) - f^*\right) + \frac{b_k^2}{2\rho\eta}\|x_0 - w^*\|^2 + \frac{\sigma^2\eta}{\rho}\sum_{t=0}^{k}\left[\gamma_t^2 b_{t+1}^2\right]$$

Proof.

Let $r_{k+1} = \|w_{k+1} - w^*\|$, then using equation 5

$$r_{k+1}^2 = \|\beta_k v_k + (1-\beta_k)\zeta_k - w^* - \gamma_k \eta \nabla f(\zeta_k, z_k)\|^2$$
$$r_{k+1}^2 = \|\beta_k v_k + (1-\beta_k)\zeta_k - w^*\|^2 + \gamma_k^2\eta^2\|\nabla f(\zeta_k, z_k)\|^2 + 2\gamma_k\eta\langle w^* - \beta_k v_k - (1-\beta_k)\zeta_k, \nabla f(\zeta_k, z_k)\rangle$$

Taking expectation wrt to z_k,

$$\begin{aligned}
\mathbb{E}[r_{k+1}^2] &= \mathbb{E}[\|\beta_k v_k + (1-\beta_k)\zeta_k - w^*\|^2] + \gamma_k^2\eta^2\mathbb{E}\|\nabla f(\zeta_k, z_k)\|^2 + 2\gamma_k\eta\,\mathbb{E}[\langle w^* - \beta_k v_k - (1-\beta_k)\zeta_k, \nabla f(\zeta_k, z_k)\rangle] \\
&\leq \|\beta_k v_k + (1-\beta_k)\zeta_k - w^*\|^2 + \gamma_k^2\eta^2\rho\|\nabla f(\zeta_k)\|^2 + 2\gamma_k\eta\langle w^* - \beta_k v_k - (1-\beta_k)\zeta_k, \nabla f(\zeta_k)\rangle + \gamma_k^2\eta^2\sigma^2 \\
&= \|\beta_k(v_k - w^*) + (1-\beta_k)(\zeta_k - w^*)\|^2 + \gamma_k^2\eta^2\rho\|\nabla f(\zeta_k)\|^2 + 2\gamma_k\eta\langle(w^* - \beta_k v_k - (1-\beta_k)\zeta_k, \nabla f(\zeta_k)\rangle + \gamma_k^2\eta^2\sigma^2 \\
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Taking expectation wrt z_k and using equations 9, 10

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\end{aligned}$$

Example - do not read!

Example - do not read!

- ⚠ Error-prone
- ⚠ Technical, lack global insights.
- ⚠ Few proof patterns.

- ? Easily fixable?
- ? Simple to adapt to variations of target inequality?
- ? Simple to adapt to algorithmic variations?

Convergence rate of a gradient step

Convergence rate of a gradient step

Toy example: What is the smallest τ such that:

$$\|x_1 - x_\star\|^2 \leqslant \tau \|x_0 - x_\star\|^2,$$

for all

- $d \in \mathbb{N}$, L-smooth and μ-strongly convex function f (notation $f \in \mathcal{F}_{\mu,L}$),
- x_0, and x_1 generated by gradient step $x_1 = x_0 - \alpha \nabla f(x_0)$,
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Computing τ?[1]

[1] Original idea from [Drori and Teboulle, 2014]. Developments here from [T, Hendrickx, Glineur, 2017].

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$$\tau = \max_{f, x_0, x_1, x_\star} \frac{\|x_1 - x_\star\|^2}{\|x_0 - x_\star\|^2}$$

$$\text{s.t. } f \in \mathcal{F}_{\mu,L}$$

Functional class

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$$\begin{aligned} \text{s.t. } & f \in \mathcal{F}_{\mu,L} && \text{Functional class} \\ & x_1 = x_0 - \alpha \nabla f(x_0) && \text{Algorithm} \\ & \nabla f(x_\star) = 0 && \text{Optimality of } x_\star \end{aligned}$$

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<u>Variables</u>: f, x_0, x_1, $x_\star$.

<u>Parameters</u>: μ, L, α.

[1] Original idea from [Drori and Teboulle, 2014]. Developments here from [T, Hendrickx, Glineur, 2017].

Sampled version

Sampled version

◇ Performance estimation problem :

$$\max_{f, x_0, x_1, x_\star} \quad \frac{\|x_1 - x_\star\|^2}{\|x_0 - x_\star\|^2}$$

$$\text{subject to} \quad f \text{ is } L\text{-smooth and } \mu\text{-strongly convex},$$

$$x_1 = x_0 - \alpha \nabla f(x_0)$$

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Sampled version

◇ Performance estimation problem **(Variables:** f**,** x_0**,** x_1**,** $x_\star$**)**:

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◇ Sampled version:

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subject to $\exists f \in \mathcal{F}_{\mu,L}$ such that $\begin{cases} f_i = f(x_i) & i = 0,\star \\ g_i = \nabla f(x_i) & i = 0,\star \end{cases}$

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◇ Sampled version: **(Variables:** x_0, x_1, $x_\star$, g_0, $g_\star$, f_0, $f_\star$**)**

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$$g_\star = 0.$$

Smooth strongly convex interpolation (or extension)

Let I index set, and associated $\{(x_i, g_i, f_i)\}_{i \in I}$: points x_i, (sub)gradients g_i and function values f_i.

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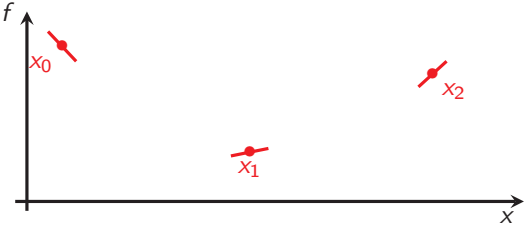

? Possible to find $f \in \mathcal{F}_{\mu,L}$ such that $f(x_i) = f_i$, and $g_i = \nabla f(x_i)$ $\forall i \in I$?

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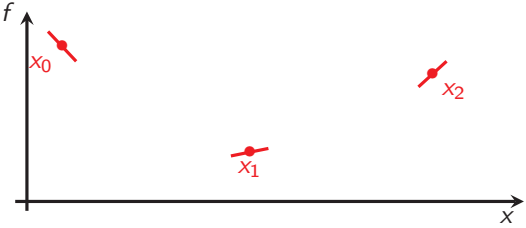

? Possible to find $f \in \mathcal{F}_{\mu,L}$ such that $f(x_i) = f_i$, and $g_i = \nabla f(x_i)\ \forall i \in I$?

- Necessary and sufficient condition: $\forall i, j \in I$

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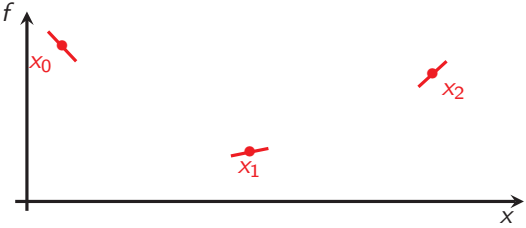
Smooth strongly convex interpolation (or extension)

Let I index set, and associated $\{(x_i, g_i, f_i)\}_{i \in I}$: points x_i, (sub)gradients g_i and function values f_i.



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- Simpler example: pick $\mu = 0$ and $L = \infty$ (just convexity):

$$f_i \geqslant f_j + \langle g_j, x_i - x_j \rangle.$$

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- Same optimal value (no relaxation): non-convex quadratic problem.

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- For $d = 1$ same as original problem by adding $\operatorname{rank}(G) \leqslant 1$.

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Fix $L = 1$, $\mu = .1$ and solve the SDP for a few values of α.

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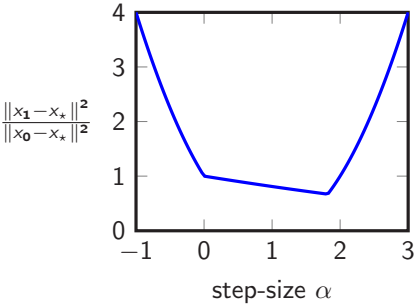

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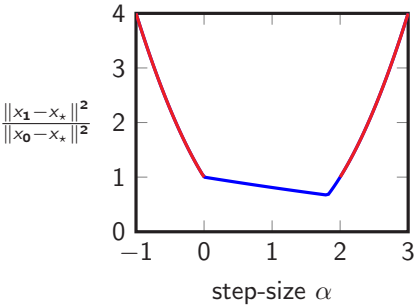

Observations:

- numerics match the known $\max\{(1-\alpha L)^2, (1-\alpha\mu)^2\}$
- recovers that gradient descent converges for $\alpha \in (0, 2/L)$ (divergence otherwise).

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- Dual problem is

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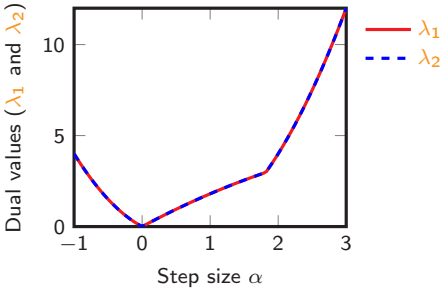

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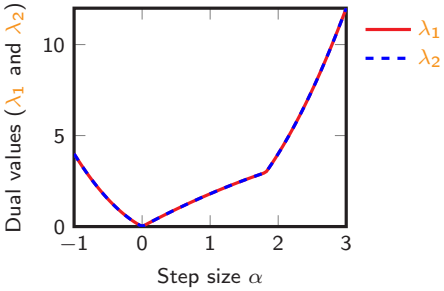

Numerics match $\lambda_1 = \lambda_2 = 2|\alpha|\rho(\alpha)$ with $\rho(\alpha) = \max\{\alpha L - 1, 1 - \alpha\mu\}$.

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- ◇ Feasible points to primal SDP correspond to lower bounds on $\tau(\mu, L, \alpha)$.
- ◇ Feasible points to dual SDP correspond to upper bounds on $\tau(\mu, L, \alpha)$.
  - – proof via linear combinations of interpolation inequalities (evaluated at $\{x_k\}_k$ and $x_\star$),
  - – proofs can be rewritten as a "sum-of-squares" certificates (sum of squared norms).

Back to legitimate questions

Legitimate questions (gradient descent, one iteration):

- ◇ anything improvable? Realistic analyses?
- ◇ How to choose the right inequalities to combine?
- ◇ Why studying this specific quantity? Possible to adapt to other quantities?
- ◇ Unique way to arrive to the desired result?
- ◇ How likely are we to find such proofs in more complicated cases?

Missing details (next)

Details that we did not cover so far:

- (convex) interpolation
- semidefinite programming,
- (Lagrange) duality.

When does the approach actually work?

- Function classes?
- Families of algorithms?
- Types of guarantees?